MH371 known-flight WSPR controls
PHaRLAP-filtered residuals: common-aircraft-altitude propagation screen

Control: −60 min

Arc-time window

Control: +60 min

2014-03-07 04:04:09 UTC
Latitude

Residual clusters: 1
Shifted-time control; aircraft distance not scored

Residual clusters: 0
Nearest plotted residual to arc-time reference: —
Time-aligned 25-km recovery: no

Withheld reference: altitude 11.6 km; time-aligned
MH371 retained ACARS workbook; linear interpolation between bracketing ACARS reports

Residual clusters: 0
Shifted-time control; aircraft distance not scored

Outside frozen 0–30°N candidate-search latitude

500 km

2014-03-07 06:11:01 UTC
Latitude

Residual clusters: 0
Shifted-time control; aircraft distance not scored

Residual clusters: 0
Nearest plotted residual to arc-time reference: —
Time-aligned 25-km recovery: no

Withheld reference: altitude 12.2 km; time-aligned
MH371 retained ACARS workbook; linear interpolation between bracketing ACARS reports

Residual clusters: 3
Shifted-time control; aircraft distance not scored

Outside frozen 0–30°N candidate-search latitude

500 km

2014-03-07 06:48:31 UTC
Latitude

Residual clusters: 0
Shifted-time control; aircraft distance not scored

Residual clusters: 0
Nearest plotted residual to arc-time reference: —
Time-aligned 25-km recovery: no

Withheld reference: altitude 12.2 km; time-aligned
MH371 retained ACARS workbook; linear interpolation between bracketing ACARS reports

Residual clusters: 6
Shifted-time control; aircraft distance not scored

Outside frozen 0–30°N candidate-search latitude

500 km

Longitude (°E)

Independent contributing links

Blind, epoch-partitioned selection; 25 km clustering, 25 km symmetric residualization, ≥3 independent links. Withheld aircraft references appear only in arc-time panels and were added after selection. Whole-window proximity uses the static arc-epoch reference; recovery uses the physical WSPR slot nearest the arc epoch and central-window truth evaluated at its midpoint. The same central-slot truth is reused as a reference in both controls; it is not aircraft truth at the shifted radio times. The orange line joins reported ACARS positions in time order in the arc-time panels only as post-selection context; positions between reports are not observed and the line does not enter selection or scoring. Latitude limits are symmetric about the withheld arc-time position for each row and identical across its three panels; pale-grey latitude bands lie outside the frozen candidate-search domain. The ruler at right is a north/south great-circle distance.

Control persistence is tested on source altitude-slice clusters across any height; only surviving sources are then collapsed into one event marker for display and counting. Marker size uses the feasible-set midrange as a geometry encoding, not evidential strength, and the audit CSV retains every source ID. Confirmatory recovery uses the surviving pre-collapse source centres and exact heights, not the displayed representative. Time-aligned recovery uses a 0.5 km primary altitude allowance because ACARS pressure altitude and modeled geometric ray height are not identical; sensitivities are 0.25, 0.5, 1 km.

BTO arc
ACARS report positions joined in time order (post-selection context)
Interpolated withheld aircraft reference
Feasible-altitude midpoint 0.5 km
Feasible-altitude midpoint 7 km
Feasible-altitude midpoint 13 km